

22MA201-ENGINEERING MATHEMATICS II

UNIT 4 & LP4- GAUSS DIVERGENCE THEOREM

Divergence theorem-Introduction

The divergence theorem is the one in which the surface integral is related to the volume integral. More precisely, the Divergence theorem relates the flux through the closed surface of a vector field to the divergence in the enclosed volume of the field. It states that the outward flux through a closed surface is equal to the integral volume within the surface of the divergence over the area.

The net flow of an area will be received by subtracting the sum of all sources by the sum of every sink. The result describes the flow by a surface of a vector and the behaviour of the vector field within it.

To state Gauss's divergence theorem in an easier way, let us break it into parts. The surface integral of a vector field over a sealed area is known as flux through the surface. Therefore, the flux through the surface is equivalent to the volume integral of the divergence over the area inside the surface. Thus, the total sum of all sources of the field in an area gives the net flux out of the area.

GAUSS DIVERGENCE THEOREM

This theorem enables us to convert a surface integral of a vector function on a closed surface into volume integral.

Statement of Gauss Divergence theorem

If V is the volume bounded by a closed surface S and if a vector function $\vec{F}$ is continuous and has continuous partial derivatives in V and on S , then

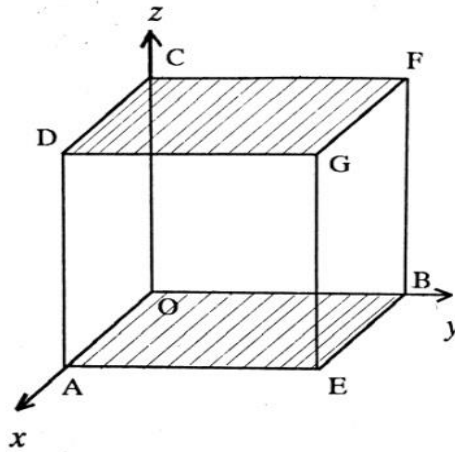
$$\iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

Where $\hat{n}$ is the unit outward normal to the surface S and $dV = dxdydz$

Problems based on gauss Divergence theorem

Example: 1. Verify the G.D.T for $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ over the cube bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.

Solution:



Gauss divergence theorem is $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$

$$\text{Given } \vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$$

$$\nabla \cdot \vec{F} = 4z - 2y + y$$

$$= 4z - y$$

$$\text{Now, R.H.S} = \iiint_V \nabla \cdot \vec{F} dv$$

$$= \int_0^1 \int_0^1 \int_0^1 (4z - y) dx dy dz$$

$$= \int_0^1 \int_0^1 [(4xz - yz)]_0^1 dy dz$$

$$= \int_0^1 \int_0^1 (4z - y) dy dz$$

$$= \int_0^1 \left(4zy - \frac{y^2}{2} \right)_0^1 dz$$

$$= \int_0^1 \left(4z - \frac{1}{2} \right) dz$$

$$= \left[4 \frac{z^2}{2} - \frac{1}{2} z \right]_0^1 = \left(2 - \frac{1}{2} \right) - 0 = \frac{3}{2}$$

$$\text{Now, L.H.S} = \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

Faces	Plane	dS	$\hat{n}$	$\vec{F} \cdot \hat{n}$	Equation	$\vec{F} \cdot \hat{n}$ on S	$= \iint_S \vec{F} \cdot \hat{n} ds$
S_1 (Bottom)	xy	$dx dy$	$-\vec{k}$	$-yz$	$z = 0$	0	$\int_0^1 \int_0^1 0 dx dy$
S_2 (Top)	xy	$dx dy$	$\vec{k}$	yz	$z = 1$	y	$\int_0^1 \int_0^1 y dx dy$

$S_3(Left)$	xz	$dxdz$	$-\vec{j}$	y^2	$y = 0$	0	$\int_0^1 \int_0^1 0 \, dxdz$
$S_4(Right)$	xz	$dxdz$	$\vec{j}$	$-y^2$	$y = 1$	-1	$\int_0^1 \int_0^1 -1 \, dxdz$
$S_5(Back)$	yz	$dydz$	$-\vec{i}$	$-4xz$	$x = 0$	0	$\int_0^1 \int_0^1 0 \, dydz$
$S_6(Front)$	yz	$dydz$	$\vec{i}$	$4xz$	$x = 1$	$4z$	$\int_0^1 \int_0^1 4z \, dydz$

$$(i) \iint_{S_1} \vec{F} \cdot \hat{n} \, ds + \iint_{S_2} \vec{F} \cdot \hat{n} \, ds = \int_0^1 \int_0^1 0 \, dxdy + \int_0^1 \int_0^1 y \, dxdy$$

$$= 0 + \int_0^1 \int_0^1 y \, dxdy$$

$$= \int_0^1 [yx]_0^1 \, dy$$

$$= \int_0^1 y \, dy$$

$$= \left[\frac{y^2}{2} \right]_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$$

$$(ii) \iint_{S_3} \vec{F} \cdot \hat{n} \, ds + \iint_{S_4} \vec{F} \cdot \hat{n} \, ds = \int_0^1 \int_0^1 0 \, dxdz + \int_0^1 \int_0^1 -1 \, dxdz$$

$$= 0 + \int_0^1 \int_0^1 -1 \, dxdz$$

$$= - \int_0^1 [x]_0^1 \, dz$$

$$= - \int_0^1 dz$$

$$= -[z]_0^1 = -[1]$$

$$(iii) \iint_{S_5} \vec{F} \cdot \hat{n} \, ds + \iint_{S_6} \vec{F} \cdot \hat{n} \, ds = \int_0^1 \int_0^1 0 \, dydz + \int_0^1 \int_0^1 4z \, dydz$$

$$= 0 + \int_0^1 \int_0^1 4z \, dydz$$

$$= \int_0^1 [4zy]_0^1 \, dz$$

$$= \int_0^1 4z \, dz$$

$$= 4 \left[\frac{z^2}{2} \right]_0^1 = 4 \left(\frac{1}{2} - 0 \right) = 2$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

$$= (i) + (ii) + (iii)$$

$$= \frac{1}{2} - 1 + 2 = \frac{3}{2}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

Hence Gauss divergence theorem is verified.

Example: 2 · Verify the G.D.T for $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$ over the rectangular parallelopiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$. (OR)

Verify the G.D.T for $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$ over the rectangular parallelopiped bounded by $x = 0, x = a, y = 0, y = b, z = 0, z = c$.

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$$

$$\nabla \cdot \vec{F} = 2x + 2y + 2z = 2(x + y + z)$$

$$\text{Now, R.H.S} = \iiint_V \nabla \cdot \vec{F} dv$$

$$= 2 \int_0^c \int_0^b \int_0^a (x + y + z) dx dy dz$$

$$= 2 \int_0^c \int_0^b \left[\left(\frac{x^2}{2} + xy + xz \right) \right]_0^a dy dz$$

$$= 2 \int_0^c \int_0^b \left(\frac{a^2}{2} + ay + az \right) dy dz$$

$$= 2 \int_0^c \left(\frac{a^2 y}{2} + \frac{ay^2}{2} + azy \right)_0^b dz$$

$$= 2 \int_0^c \left(\frac{a^2 b}{2} + \frac{ab^2}{2} + azb \right) dz$$

$$= 2 \left[\frac{a^2 bz}{2} + \frac{ab^2 z}{2} + \frac{abz^2}{2} \right]_0^c$$

$$= 2 \left(\frac{a^2 bc}{2} + \frac{ab^2 c}{2} + \frac{abc^2}{2} \right)$$

$$= abc(a + b + c)$$

$$\text{Now, L.H.S} = \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

Faces	Plane	dS	$\hat{n}$	$\vec{F} \cdot \hat{n}$	Eqn	$\vec{F} \cdot \hat{n}$ on S	$= \iint_S \vec{F} \cdot \hat{n} ds$
S_1 (Bottom)	xy	$dxdy$	$-\vec{k}$	$-(z^2 - xy)$	$z = 0$	xy	$\int_0^b \int_0^a xy dx dy$
S_2 (Top)	xy	$dxdy$	$\vec{k}$	$(z^2 - xy)$	$z = c$	$c^2 - xy$	$\int_0^b \int_0^a c^2 - xy dx dy$
S_3 (Left)	xz	$dx dz$	$-\vec{j}$	$-(y^2 - xz)$	$y = 0$	xz	$\int_0^c \int_0^a xz dx dz$
S_4 (Right)	xz	$dx dz$	$\vec{j}$	$(y^2 - xz)$	$y = b$	$b^2 - xz$	$\int_0^c \int_0^a b^2 - xz dx dz$

$S_5(Back)$	yz	$dydz$	$-\vec{i}$	$-(x^2 - yz)$	$x = 0$	yz	$\int_0^c \int_0^b yz \, dydz$
$S_6(Front)$	yz	$dydz$	$\vec{i}$	$(x^2 - yz)$	$x = a$	$a^2 - yz$	$\int_0^c \int_0^b a^2 - yz \, dydz$

$$(i) \iint_{S_1} \vec{F} \cdot \hat{n} \, ds + \iint_{S_2} \vec{F} \cdot \hat{n} \, ds = \int_0^b \int_0^a xy \, dx dy + \int_0^b \int_0^a c^2 - xy \, dx dy$$

$$= \int_0^b \int_0^a c^2 \, dx dy$$

$$= c^2 \int_0^a dx \int_0^b dy$$

$$= c^2 [x]_0^a [y]_0^b = c^2 ab$$

$$(ii) \iint_{S_3} \vec{F} \cdot \hat{n} \, ds + \iint_{S_4} \vec{F} \cdot \hat{n} \, ds = \int_0^c \int_0^a xz \, dx dz + \int_0^c \int_0^a b^2 - xz \, dx dz$$

$$= \int_0^c \int_0^a b^2 \, dx dz$$

$$= b^2 \int_0^a dx \int_0^c dz$$

$$= b^2 [x]_0^a [z]_0^c = b^2 ac$$

$$(iii) \iint_{S_5} \vec{F} \cdot \hat{n} \, ds + \iint_{S_6} \vec{F} \cdot \hat{n} \, ds = \int_0^c \int_0^b yz \, dy dz + \int_0^c \int_0^b a^2 - yz \, dy dz$$

$$= \int_0^c \int_0^b a^2 \, dy dz$$

$$= a^2 \int_0^b dy \int_0^c dz$$

$$= a^2 [y]_0^b [z]_0^c = a^2 bc$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

$$= (i) + (ii) + (iii)$$

$$= abc(a + b + c)$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

Hence Gauss divergence theorem is verified.

Example: 3. Verify divergence theorem for $\vec{F} = (2x - z)\vec{i} + x^2y\vec{j} - xz^2\vec{k}$ over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

Solution:

Gauss divergence theorem is $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$

$$\text{Given } \vec{F} = (2x - z)\vec{i} + x^2y\vec{j} - xz^2\vec{k}$$

$$\nabla \cdot \vec{F} = 2 + x^2 - 2xz$$

$$\text{Now, R.H.S} = \iiint_V \nabla \cdot \vec{F} dv$$

$$= \int_0^1 \int_0^1 \int_0^1 (2 + x^2 - 2xz) dx dy dz$$

$$= \int_0^1 \int_0^1 \left[\left(2x + \frac{x^3}{3} - \frac{2zx^2}{2} \right) \right]_0^1 dy dz$$

$$= \int_0^1 \int_0^1 \left(2 + \frac{1}{3} - z \right) dy dz$$

$$= \int_0^1 \left(2y + \frac{1}{3}y - zy \right)_0^1 dz$$

$$= \int_0^1 \left(2 + \frac{1}{3} - z \right) dz$$

$$= \left[2z + \frac{1}{3}z - \frac{z^2}{2} \right]_0^1$$

$$= \left(2 + \frac{1}{3} - \frac{1}{2} \right) - 0 = \frac{11}{6}$$

$$\text{Now, L.H.S} = \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

Faces	Plane	dS	$\hat{n}$	$\vec{F} \cdot \hat{n}$	Equation	$\vec{F} \cdot \hat{n}$ on S	$= \iint_S \vec{F} \cdot \hat{n} ds$
S_1 (Bottom)	xy	$dxdy$	$-\vec{k}$	xz^2	$z = 0$	0	$\int_0^1 \int_0^1 0 dx dy$
S_2 (Top)	xy	$dxdy$	$\vec{k}$	$-xz^2$	$z = 1$	$-x$	$\int_0^1 \int_0^1 (-x) dx dy$
S_3 (Left)	xz	$dxdz$	$-\vec{j}$	$-x^2y$	$y = 0$	0	$\int_0^1 \int_0^1 0 dx dz$
S_4 (Right)	xz	$dxdz$	$\vec{j}$	x^2y	$y = 1$	x^2	$\int_0^1 \int_0^1 x^2 dx dz$
S_5 (Back)	yz	$dydz$	$-\vec{i}$	$-(2x - z)$	$x = 0$	z	$\int_0^1 \int_0^1 z dy dz$
S_6 (Front)	yz	$dydz$	$\vec{i}$	$(2x - z)$	$x = 1$	$2 - z$	$\int_0^1 \int_0^1 2 - z dy dz$

$$(i) \iint_{S_1} \vec{F} \cdot \hat{n} ds + \iint_{S_2} \vec{F} \cdot \hat{n} ds = \int_0^1 \int_0^1 0 dx dy + \int_0^1 \int_0^1 (-x) dx dy$$

$$= \int_0^1 \int_0^1 (-x) dx dy$$

$$= - \int_0^1 \left[\frac{x^2}{2} \right]_0^1 dy$$

$$= - \int_0^1 \frac{1}{2} dy$$

$$= - \left[\frac{1}{2} y \right]_0^1 = - \left(\frac{1}{2} - 0 \right) = -\frac{1}{2}$$

$$(ii) \iint_{S_3} \vec{F} \cdot \hat{n} ds + \iint_{S_4} \vec{F} \cdot \hat{n} ds = \int_0^1 \int_0^1 0 dx dz + \int_0^1 \int_0^1 x^2 dx dz$$

$$= \int_0^1 \int_0^1 x^2 dx dz$$

$$= \int_0^1 \left[\frac{x^3}{3} \right]_0^1 dz$$

$$= \int_0^1 \frac{1}{3} dz$$

$$= \left[\frac{1}{3} z \right]_0^1 = \left(\frac{1}{3} - 0 \right) = \frac{1}{3}$$

$$(iii) \iint_{S_5} \vec{F} \cdot \hat{n} ds + \iint_{S_6} \vec{F} \cdot \hat{n} ds = \int_0^1 \int_0^1 z dy dz + \int_0^1 \int_0^1 (2-z) dy dz$$

$$= \int_0^1 \int_0^1 2 dy dz$$

$$= 2 \int_0^1 [y]_0^1 dz$$

$$= 2 \int_0^1 dz$$

$$= 2 [z]_0^1 = 2$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

$$= (i) + (ii) + (iii)$$

$$= -\frac{1}{2} + \frac{1}{3} + 2 = \frac{11}{6}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

Hence Gauss divergence theorem is verified.

Example: 4. Verify divergence theorem for $\vec{F} = x^2\vec{i} + z\vec{j} + yz\vec{k}$ over the cube bounded by $x = \pm 1$, $y = \pm 1$, $z = \pm 1$.

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = x^2\vec{i} + z\vec{j} + yz\vec{k}$$

$$\nabla \cdot \vec{F} = 2x + y$$

$$\text{Now, R.H.S} = \iiint_V \nabla \cdot \vec{F} dv$$

$$= \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (2x + y) dx dy dz$$

$$\begin{aligned}
&= \int_{-1}^1 \int_{-1}^1 \left[\left(2 \frac{x^2}{2} + yx \right) \right]_{-1}^1 dydz \\
&= \int_{-1}^1 \int_{-1}^1 [(1+y) - (1-y)] dydz \\
&= \int_{-1}^1 \int_{-1}^1 [2y] dydz \\
&= \int_{-1}^1 \left(2 \frac{y^2}{2} \right)_{-1}^1 dz \\
&= \int_{-1}^1 [(1) - ((-1)^2)] dz \\
&= \int_{-1}^1 [0] dz \\
&= 0
\end{aligned}$$

$$\text{Now, L.H.S} = \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

Faces	Plane	dS	$\hat{n}$	$\vec{F} \cdot \hat{n}$	Equation	$\vec{F} \cdot \hat{n}$ on S	$= \iint_S \vec{F} \cdot \hat{n} ds$
$S_1(\text{Bottom})$	xy	$dxdy$	$-\vec{k}$	$-yz$	$z = -1$	y	$\int_{-1}^1 \int_{-1}^1 y dxdy$
$S_2(\text{Top})$	xy	$dxdy$	$\vec{k}$	yz	$z = 1$	y	$\int_{-1}^1 \int_{-1}^1 y dxdy$
$S_3(\text{Left})$	xz	$dxdz$	$-\vec{j}$	$-z$	$y = -1$	$-z$	$\int_{-1}^1 \int_{-1}^1 -z dxdz$
$S_4(\text{Right})$	xz	$dxdz$	$\vec{j}$	z	$y = 1$	z	$\int_{-1}^1 \int_{-1}^1 z dxdz$
$S_5(\text{Back})$	yz	$dydz$	$-\vec{i}$	$-x^2$	$x = -1$	-1	$\int_{-1}^1 \int_{-1}^1 -1 dydz$
$S_6(\text{Front})$	yz	$dydz$	$\vec{i}$	x^2	$x = 1$	1	$\int_{-1}^1 \int_{-1}^1 dydz$

$$\begin{aligned}
(i) \iint_{S_1} \vec{F} \cdot \hat{n} ds + \iint_{S_2} \vec{F} \cdot \hat{n} ds &= \int_{-1}^1 \int_{-1}^1 y dxdy + \int_{-1}^1 \int_{-1}^1 y dxdy \\
&= \int_{-1}^1 \int_{-1}^1 2y dxdy \\
&= 2 \int_{-1}^1 [xy]_{-1}^1 dy \\
&= 2 \int_{-1}^1 [(y) - (-y)] dy \\
&= 2 \int_{-1}^1 2y dy \\
&= 4 \left[\frac{y^2}{2} \right]_{-1}^1 = 4 \left[\left(\frac{1}{2} \right) - \left(\frac{1}{2} \right) \right] = 0
\end{aligned}$$

$$\begin{aligned}
(ii) \iint_{S_3} \vec{F} \cdot \hat{n} ds + \iint_{S_4} \vec{F} \cdot \hat{n} ds &= \int_{-1}^1 \int_{-1}^1 -z dxdz + \int_{-1}^1 \int_{-1}^1 z dxdz \\
&= \int_{-1}^1 \int_{-1}^1 0 dxdz
\end{aligned}$$

$$= 0$$

$$(iii) \iint_{S_5} \vec{F} \cdot \hat{n} ds + \iint_{S_6} \vec{F} \cdot \hat{n} ds = - \int_{-1}^1 \int_{-1}^1 dx dz + \int_{-1}^1 \int_{-1}^1 dx dz$$

$$= 0$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iint_{S_1} + \iint_{S_2} + \iint_{S_3} + \iint_{S_4} + \iint_{S_5} + \iint_{S_6}$$

$$= (i) + (ii) + (iii)$$

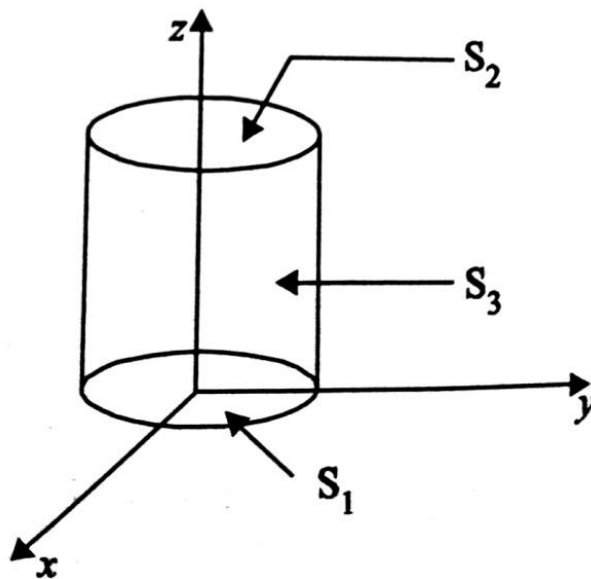
$$= 0$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

Hence , Gauss divergence theorem is verified.

Example: 5. Verify divergence theorem for the function $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ taken over the surface bounded by the cylinder $x^2 + y^2 = 4$ and $z = 0, z = 3$.

Solution:



$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$$

$$\nabla \cdot \vec{F} = 4 - 4y + 2z$$

Limits:

$$z = 0 \text{ to } 3$$

$$x^2 + y^2 = 4 \Rightarrow y^2 = 4 - x^2$$

$$\Rightarrow y = \pm\sqrt{4 - x^2}$$

$$\therefore y = -\sqrt{4 - x^2} \text{ to } \sqrt{4 - x^2}$$

$$\text{Put } y = 0 \Rightarrow x^2 = 4$$

$$\Rightarrow x = \pm 2$$

$$\therefore y = -2 \text{ to } 2$$

$$\therefore \text{R.H.S} = \iiint_V \nabla \cdot \vec{F} \, dv$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) \, dz dy dx$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left[4z - 4yz + 2\frac{z^2}{2} \right]_0^3 dy dx$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (12 - 12y + 9) dy dx$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y) dy dx$$

$$= 2 \int_{-2}^2 \int_0^{\sqrt{4-x^2}} 21 \, dy dx$$

$$\left[\because \int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx \text{ if } f(x) \text{ is even} \right. \\ \left. = 0 \text{ if } f(x) \text{ is odd} \right]$$

$$= 42 \int_{-2}^2 [y]_0^{\sqrt{4-x^2}} dx$$

$$= 42 \int_{-2}^2 \sqrt{4-x^2} \, dx$$

$$= 42 \times 2 \int_0^2 \sqrt{4-x^2} \, dx$$

$$[\because \text{even function}]$$

$$= 84 \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2$$

$$= 84 [0 + 2 \sin^{-1}(1)]$$

$$= 84 \left[2 \times \frac{\pi}{2} \right]$$

$$= 84 \pi$$

$$\text{L.H.S} = \iint_S \vec{F} \cdot \hat{n} \, ds$$

$$= \iint_{S_1} + \iint_{S_2} + \iint_{S_3}$$

Along S_1 (bottom):

$$xy \text{ -plane} \Rightarrow z = 0, dz = 0$$

$$\text{And } ds = dxdy, \hat{n} = -\vec{k}$$

$$\therefore \vec{F} \cdot \hat{n} = (4x \vec{i} - 2y^2 \vec{j} + z^2 \vec{k}) \cdot (-\vec{k})$$

$$= -z^2 = 0$$

$$\therefore \iint_{S_1} \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} 0 = 0$$

Along S_2 (top):

$$xy \text{ -plane} \Rightarrow z = 3, dz = 0$$

$$\text{And } ds = dxdy, \hat{n} = \vec{k}$$

$$\therefore \vec{F} \cdot \hat{n} = (4x \vec{i} - 2y^2 \vec{j} + z^2 \vec{k}) \cdot (\vec{k})$$

$$= z^2 = 9$$

$$\begin{aligned}\therefore \iint_{S_2} \vec{F} \cdot \hat{n} ds &= \iint_{S_2} 9 dx dy \\ &= \iint_R 9 dx dy \\ &= 9 (\text{Area of the circle}) \\ &= 9 (\pi r^2) \quad [\because r = 2] \\ &= 36 \pi\end{aligned}$$

Along S_3 (curved surface):

$$\text{Given } x^2 + y^2 = 4$$

$$\text{Let } \varphi = x^2 + y^2 - 4$$

$$\nabla \varphi = \vec{i} \frac{\partial \varphi}{\partial x} + \vec{j} \frac{\partial \varphi}{\partial y} + \vec{k} \frac{\partial \varphi}{\partial z}$$

$$= 2x\vec{i} + 2y\vec{j}$$

$$|\nabla \varphi| = \sqrt{4x^2 + 4y^2} = 2\sqrt{4} = 4$$

$$\begin{aligned}\hat{n} &= \frac{\nabla \varphi}{|\nabla \varphi|} = \frac{2(x\vec{i} + y\vec{j})}{4} \\ &= \frac{x\vec{i} + y\vec{j}}{2}\end{aligned}$$

The cylindrical coordinates are

$$x = 2 \cos \theta, \quad y = 2 \sin \theta \quad ds = 2 dz d\theta$$

Where z varies from 0 to 3

θ varies from 0 to 2π

$$\text{Now } \vec{F} \cdot \hat{n} = (4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}) \cdot \left(\frac{x\vec{i} + y\vec{j}}{2}\right)$$

$$= 2x^2 - y^3$$

$$= 2(2 \cos \theta)^2 - (2 \sin \theta)^3$$

$$= 8 \cos^2 \theta - 8 \sin^3 \theta$$

$$= 8 \left[\frac{1 + \cos 2\theta}{2} - \left(\frac{3 \sin \theta - \sin 3\theta}{4} \right) \right]$$

$$\therefore \iint_{S_3} \vec{F} \cdot \hat{n} ds = 8 \int_0^{2\pi} \int_0^3 \left(\frac{1}{2} + \frac{\cos 2\theta}{2} - \frac{3 \sin \theta}{4} + \frac{\sin 3\theta}{4} \right) 2 dz d\theta$$

$$= 16 \int_0^{2\pi} \left(\frac{1}{2} + \frac{\cos 2\theta}{2} - \frac{3 \sin \theta}{4} + \frac{\sin 3\theta}{4} \right) [z]_0^3 d\theta$$

$$= 48 \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} - \frac{3 \cos \theta}{4} - \frac{\cos 3\theta}{12} \right]_0^{2\pi}$$

$$= 48 \left[\left(\frac{2\pi}{2} + \frac{3}{4} - \frac{1}{12} \right) - \left(\frac{3}{4} - \frac{1}{12} \right) \right]$$

$$= 48 \pi$$

$$\begin{aligned}\text{L.H.S} &= \iint_S \vec{F} \cdot \hat{n} ds = 0 + 36\pi + 48\pi \\ &= 84\pi\end{aligned}$$

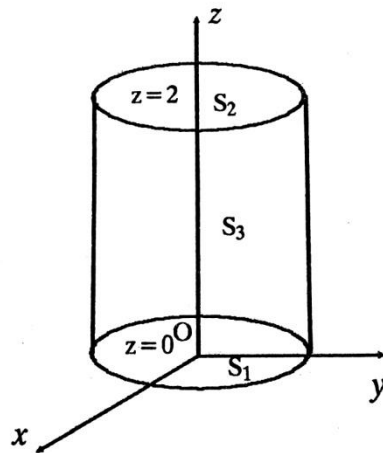
$$\therefore \text{L.H.S} = \text{R.H.S}$$

$$(i.e) \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

Hence Gauss divergence theorem is verified.

Example: 6. Verify divergence theorem for the function $\vec{F} = y\vec{i} + x\vec{j} + z^2\vec{k}$ over the cylindrical region bounded by $x^2 + y^2 = 9$ and $z = 0, z = 2$.

Solution:



$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = y\vec{i} + x\vec{j} + z^2\vec{k}$$

$$\nabla \cdot \vec{F} = 2z$$

Limits:

$$z = 0 \text{ to } 2$$

$$x^2 + y^2 = 9 \Rightarrow y^2 = 9 - x^2$$

$$\Rightarrow y = \pm\sqrt{9 - x^2}$$

$$\therefore y = -\sqrt{9 - x^2} \text{ to } \sqrt{9 - x^2}$$

$$\text{Put } y = 0 \Rightarrow x^2 = 9$$

$$\Rightarrow x = \pm 3$$

$$\therefore y = -3 \text{ to } 3$$

$$\therefore \text{R.H.S} = \iiint_V \nabla \cdot \vec{F} dv$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^2 (2z) dz dy dx$$

$$= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \left[2 \frac{z^2}{2} \right]_0^2 dy dx$$

$$\begin{aligned}
&= 4 \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dy dx \\
&= 4 \int_{-3}^3 [y]_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx \\
&= 4 \int_{-3}^3 2\sqrt{9-x^2} dx \\
&= 8 \times 2 \int_0^3 \sqrt{9-x^2} dx \quad [\because \text{even function}] \\
&= 16 \left[\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \frac{x}{3} \right]_0^3 \\
&= 16 \left[0 + \frac{9}{2} \sin^{-1}(1) \right] \\
&= 16 \left[\frac{9}{2} \times \frac{\pi}{2} \right] \\
&= 36\pi
\end{aligned}$$

$$\text{L.H.S} = \iint_S \vec{F} \cdot \hat{n} ds$$

$$= \iint_{S_1} + \iint_{S_2} + \iint_{S_3}$$

Along S_1 (bottom):

$$xy \text{ -plane} \Rightarrow z = 0, dz = 0$$

$$\text{And } ds = dxdy, \hat{n} = -\vec{k}$$

$$\begin{aligned}
\therefore \vec{F} \cdot \hat{n} &= (y\vec{i} + x\vec{j} + z^2\vec{k}) \cdot (-\vec{k}) \\
&= -z^2 = 0
\end{aligned}$$

$$\therefore \iint_{S_1} \vec{F} \cdot \hat{n} ds = \iint_{S_1} 0 = 0$$

Along S_2 (top):

$$xy \text{ -plane} \Rightarrow z = 2, dz = 0$$

$$\text{And } ds = dxdy, \hat{n} = \vec{k}$$

$$\begin{aligned}
\therefore \vec{F} \cdot \hat{n} &= (y\vec{i} + x\vec{j} + z^2\vec{k}) \cdot (\vec{k}) \\
&= z^2 = 4
\end{aligned}$$

$$\therefore \iint_{S_2} \vec{F} \cdot \hat{n} ds = \iint_{S_2} 4 dxdy$$

$$= \iint_R 4 dxdy$$

$$= 4 (\text{Area of the circle})$$

$$= 4 (\pi r^2) \quad [\because r = 2]$$

$$= 36\pi$$

Along S_3 (curved surface):

$$\text{Given } x^2 + y^2 = 9$$

$$\text{Let } \varphi = x^2 + y^2 - 9$$

$$\nabla \varphi = \vec{i} \frac{\partial \varphi}{\partial x} + \vec{j} \frac{\partial \varphi}{\partial y} + \vec{k} \frac{\partial \varphi}{\partial z}$$

$$= 2x\vec{i} + 2y\vec{j}$$

$$|\nabla \varphi| = \sqrt{4x^2 + 4y^2} = 2\sqrt{9} = 6$$

$$\begin{aligned} \hat{n} &= \frac{\nabla \varphi}{|\nabla \varphi|} = \frac{2(x\vec{i} + y\vec{j})}{6} \\ &= \frac{x\vec{i} + y\vec{j}}{3} \end{aligned}$$

The cylindrical coordinates are

$$x = 3 \cos \theta, \quad y = 3 \sin \theta$$

$$ds = 3dzd\theta$$

Where z varies from 0 to 2

θ varies from 0 to 2π

$$\begin{aligned} \text{Now } \vec{F} \cdot \hat{n} &= (y\vec{i} + x\vec{j} + z^2\vec{k}) \cdot \left(\frac{x\vec{i} + y\vec{j}}{3}\right) \\ &= \frac{xy}{3} + \frac{xy}{3} = \frac{2xy}{3} \\ &= \frac{2}{3} (3 \cos \theta) (3 \sin \theta) \\ &= 2 \times 3 \sin \theta \cos \theta \\ &= 3 \sin 2\theta \end{aligned}$$

$$\begin{aligned} \therefore \iint_{S_3} \vec{F} \cdot \hat{n} \, ds &= 3 \int_0^{2\pi} \int_0^2 (\sin 2\theta) 3dzd\theta \\ &= 9 \int_0^{2\pi} (\sin 2\theta) [z]_0^2 d\theta \\ &= 9 \left[-\frac{\cos 2\theta}{2} \right]_0^{2\pi} \\ &= -9 [1 - 1] \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{L.H.S} &= \iint_S \vec{F} \cdot \hat{n} \, ds = 0 + 36\pi + 0 \\ &= 36\pi \end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

$$(i.e) \iint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V \nabla \cdot \vec{F} \, dv$$

Hence Gauss divergence theorem is verified.

Example: 7. If S is any closed surface enclosing a volume V and if $\vec{F} = ax\vec{i} + by\vec{j} + cz\vec{k}$, prove that $\iint_S \vec{F} \cdot \hat{n} ds = (a + b + c)V$. Deduce that $\iint_S \vec{F} \cdot \hat{n} ds = \frac{4\pi}{3}(a + b + c)$ if S is the surface of the sphere $x^2 + y^2 + z^2 = 1$.

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = ax\vec{i} + by\vec{j} + cz\vec{k}$$

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = a + b + c$$

$$\begin{aligned} \therefore \iint_S \vec{F} \cdot \hat{n} ds &= \iiint_V (a + b + c) dv \\ &= (a + b + c)V \end{aligned}$$

If S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ then $V = \frac{4}{3}\pi(1)^3 = \frac{4\pi}{3}$

$$\begin{aligned} \therefore \iint_S \vec{F} \cdot \hat{n} ds &= (a + b + c) \frac{4\pi}{3} \\ &= \frac{4\pi}{3} (a + b + c) \end{aligned}$$

Example: Using the divergence theorem of Gauss evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$, and S is the sphere $x^2 + y^2 + z^2 = a^2$.

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\text{Given } \vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$$

$$\begin{aligned} \nabla \cdot \vec{F} &= 3x^2 + 3y^2 + 3z^2 \\ &= 3(x^2 + y^2 + z^2) \end{aligned}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = 3 \iiint_V (x^2 + y^2 + z^2) dxdydz$$

Here we have to use spherical polar co – ordinates.

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$

$$x^2 + y^2 + z^2 = a^2 \quad \text{and} \quad dxdydz = r^2 \sin \theta dr d\theta d\varphi$$

$$\therefore 3 \iiint_V (x^2 + y^2 + z^2) dxdydz = 3 \int_0^{2\pi} \int_0^\pi \int_0^a r^2 r^2 \sin \theta dr d\theta d\varphi$$

$$= 3 \int_0^{2\pi} \int_0^\pi \left[\frac{r^5}{5} \sin \theta \right]_0^a d\theta d\varphi$$

$$= \frac{3a^5}{5} \int_0^{2\pi} [-\cos \theta]_0^\pi d\varphi$$

$$\begin{aligned}
&= \frac{3a^5}{5} \int_0^{2\pi} (-\cos \pi + \cos 0) d\varphi \\
&= \frac{6a^5}{5} [\varphi]_0^{2\pi} \\
&= \frac{6a^5}{5} (2\pi) = \frac{12\pi a^5}{5}
\end{aligned}$$

Example: 8. Show that $\iint_S \text{curl } \vec{F} \cdot \hat{n} ds = 0$ where S is any closed surface.

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot (\nabla \times \vec{F}) dv$$

where V is the volume of the closed surface S.

$$\text{Since } \nabla \cdot (\nabla \times \vec{F}) = 0, \text{ we get } \iiint_V \nabla \cdot (\nabla \times \vec{F}) dv = 0$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dv = 0 \text{ (or) } \iint_S \text{curl } \vec{F} \cdot \hat{n} ds = 0$$

Example: 9. Prove that $\iint_S \frac{\vec{r} \cdot \hat{n}}{r^2} ds = \iiint_V \frac{dv}{r^2}$

Solution:

$$\text{Gauss divergence theorem is } \iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$$

$$\therefore \iint_S \frac{\vec{r}}{r^2} \cdot \hat{n} ds = \iiint_V \nabla \cdot \left(\frac{\vec{r}}{r^2} \right) dv$$

$$\begin{aligned}
\text{Now } \nabla \cdot \frac{\vec{r}}{r^2} &= \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot \left(\frac{x\vec{i} + y\vec{j} + z\vec{k}}{r^2} \right) \\
&= \frac{\partial}{\partial x} \left(\frac{x}{r^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^2} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^2} \right) \\
&= \sum \frac{r^2(1) - x \cdot 2r \frac{\partial r}{\partial x}}{r^4} \\
&= \sum \frac{r^2 - 2xr \left(\frac{x}{r} \right)}{r^4} \\
&= \sum \frac{r^2 - 2x^2}{r^4} \\
&= \frac{3r^2 - 2(x^2 + y^2 + z^2)}{r^4} \\
&= \frac{3r^2 - 2r^2}{r^4} = \frac{r^2}{r^4} = \frac{1}{r^2}
\end{aligned}$$

$$\therefore \iint_S \frac{\vec{r}}{r^2} \cdot \hat{n} ds = \iiint_V \frac{1}{r^2} dv$$

Example: 10. Evaluate $\iint_S \vec{r} \cdot \hat{n} ds$ where S is a closed surface using Gauss divergence theorem.

Solution:

Gauss divergence theorem is $\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \nabla \cdot \vec{F} dv$

$$\begin{aligned}
 \therefore \iint_S \vec{r} \cdot \hat{n} ds &= \iiint_V (\nabla \cdot \vec{r}) dv \\
 &= \iiint_V [\nabla \cdot (x\vec{i} + y\vec{j} + z\vec{k})] dv \\
 &= \iiint_V (1 + 1 + 1) dv \\
 &= 3 \iiint_V dv \\
 &= 3V
 \end{aligned}$$

Exercise

1. Verify divergence theorem for the function $\vec{F} = (x^2 - yz)\vec{i} - (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ over the surface bounded by $x = 0, x = 1, y = 0, y = 2, z = 0, z = 3$ **Ans:** 36

2. Verify divergence theorem for the function $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ over the cube $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ **Ans:** Common value = $\frac{3}{2}$

3. Verify divergence theorem for the function $\vec{F} = (2x - z)\vec{i} - x^2y\vec{j} - xz^2\vec{k}$ over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ **Ans:** Common value = $\frac{11}{6}$

4. Verify divergence theorem for $\vec{F} = xy^2\vec{i} + yz^2\vec{j} + zx^2\vec{k}$ over the region $x^2 + y^2 = 4$ and $z = 0, z = 3$ **Ans:** Common value = 84π

Discussion Questions:

1. What is the difference between Stokes and Gauss theorem?
2. What is the importance of the Gauss Divergence Theorem?
3. What does the Gauss divergence theorem relate to?